SONiX 32-Bit Cortex-M0 Micro-Controller

DMA Memory Example Code

AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/27	1. First version.

Table of Content

1 OVERVIEW

This document provides our customers the C program examples of DMA Memory-to-memory transfer mode.

2 FEATURE

- DMA FLASH to SRAM transfer example.
- DMA SRAM to SRAM transfer example.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Initialization

Initial DMA function. User can modify the settings of DMA in these function. The setting of DMA should refer to the table in the datasheet.

```
// Init DMA
DMAMtoMInit();
```

```
void DMAMtoMInit(void)
{
  DMA_InitSt stDMACH_Init;

  // Set DMA CH0 as M(SRAM/FLASH) to M(SRAM/FLASH) mode
  // Source : Memory, Increment, 8bit, no request, Burst 1
  stDMACH_Init.b_SrcAddrCtrl = DMA_SRCAD_CTL_INC;
  stDMACH_Init.b_SrcMode     = DMA_SRC_MEMORY;
  stDMACH_Init.b_SrcWidth    = DMA_SRC_WIDTH_8BIT;
  stDMACH_Init.b_SrcReqSel   = DMA_SRCRS_NONE;
  stDMACH_Init.b_BurstSize   = DMA_BURST_1;
  // Destination : Memory, Fixed, 8bit, no request
  stDMACH_Init.b_DstAddrCtrl = DMA_DSTAD_CTL_INC;
  stDMACH_Init.b_DstMode     = DMA_DST_MEMORY;
  stDMACH_Init.b_DstWidth    = DMA_DST_WIDTH_8BIT;
  stDMACH_Init.b_DstReqSel   = DMA_DSTRS_NONE;
  // Channel : LV0, Enable TC/ABT interrupt
  stDMACH_Init.b_Priority    = DMA_CHPRI_LV0;
  stDMACH_Init.b_IntTCEn     = DMA_INT_TC_MSK_DIS;
  stDMACH_Init.b_IntABTEn    = DMA_INT_ABT_MSK_DIS;

  // Initial DMA
  DMA_Init(&stDMACH_Init, eDMA_CH0);
}
```

4.2 DMA FLASH to SRAM transfer

The source data are 256 bytes of random data in FLASH. The destination address is SRAM.

```
// Random Flash data
const uint8_t b_aryFlashSrcData[DMA_TEST_SIZE] =
{
  0x59, 0x79, 0x69, 0xfd, 0x9d, 0xe3, 0xf0, 0xef, 0x2d, 0xed, 0xb4, 0xb4, 0x05, 0x8b, 0xff, 0x08,
  0x19, 0xe5, 0x47, 0x2f, 0x2b, 0x33, 0x49, 0x30, 0x82, 0x04, 0x62, 0x9a, 0xdd, 0xec, 0xa3, 0x3b,
  0xc5, 0xf9, 0x85, 0x38, 0x54, 0x8f, 0x5d, 0xa0, 0x0f, 0x19, 0x07, 0x0b, 0xdd, 0xb0, 0xea, 0x33,
  0xb9, 0x77, 0x25, 0x10, 0xc6, 0x84, 0x6c, 0xec, 0x32, 0xb5, 0x94, 0x70, 0x4d, 0x68, 0xa3, 0x33,
  0x81, 0x4c, 0x33, 0x20, 0x3b, 0x10, 0xf6, 0x78, 0x90, 0x79, 0x13, 0xaf, 0xe5, 0xe0, 0x83, 0x02,
  0xe9, 0xad, 0xec, 0x63, 0x5f, 0xc5, 0x6e, 0xd1, 0xaa, 0x6b, 0xc5, 0xaf, 0xf9, 0x8c, 0x84, 0x8f,
  0x27, 0xf3, 0xb3, 0xe4, 0xb7, 0xf0, 0xc2, 0x14, 0x7e, 0x37, 0xa1, 0xc9, 0xa0, 0xf3, 0xed, 0x2e,
  0x50, 0x17, 0xa7, 0xbe, 0x65, 0x98, 0xeb, 0xa2, 0x6a, 0x7d, 0xdf, 0x06, 0x69, 0xe1, 0x5e, 0x17,
  0xd7, 0x31, 0xdc, 0x00, 0xb3, 0xa3, 0x7c, 0x20, 0x3b, 0x3d, 0x91, 0xad, 0x95, 0x3c, 0x57, 0xbf,
  0x18, 0x39, 0x82, 0xed, 0x90, 0x10, 0xec, 0x4d, 0xaf, 0x1d, 0x5c, 0xd0, 0x84, 0x67, 0xe6, 0x90,
  0xff, 0x9c, 0xa9, 0xf3, 0xa0, 0xc6, 0xaa, 0x39, 0x44, 0xa5, 0x79, 0x01, 0xab, 0x35, 0x38, 0x6d,
  0xe5, 0x54, 0xf0, 0xc5, 0x43, 0xd6, 0x44, 0xb5, 0x1d, 0xc5, 0x50, 0x7b, 0x3b, 0xfd, 0xa2, 0xfb,
  0x60, 0xae, 0xa4, 0x3b, 0xa5, 0xe5, 0x64, 0xc3, 0xbc, 0x02, 0x14, 0xff, 0xe2, 0xd5, 0xe1, 0x44,
  0x65, 0x94, 0x04, 0x0b, 0x5d, 0xc9, 0xbd, 0xb3, 0xbd, 0x4d, 0x10, 0xff, 0x54, 0x01, 0x80, 0x2b,
  0xb5, 0x25, 0xf3, 0x04, 0xb0, 0x97, 0x01, 0x09, 0xdd, 0xf9, 0x5b, 0x11, 0xeb, 0x31, 0x0a, 0x4a,
  0x60, 0x10, 0x20, 0xca, 0x07, 0x6c, 0x39, 0x8b, 0x72, 0xb2, 0x79, 0xa4, 0x4d, 0x2a, 0xb9, 0x3d
};
```

After DMA is enabled, DMA starts transmitting from FLASH address to SRAM address. The DMA TC flag and interrupt will be set when all data transfers are completed.

```
/* Example 1: DMA Flash to SRAM Start*/

// Start DMA transfer
DMAFlashtoSramStart();

// Wait for DMA TC interrupt flag
while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);

// Check SRAM data = FLASH data
for(i = 0; i<DMA_TEST_SIZE ;i++)
{
  if(b_aryFlashSrcData[i] != b_arySramDstData[i])
    while(1); //Fail
}

/* Example 1: DMA Flash to SRAM End*/
```

4.3 DMA SRAM to SRAM transfer

The source data are 256 bytes of data in SRAM. The destination address is SRAM.

```
// Init data buffer
for(i = 0; i<DMA_TEST_SIZE ;i++)
{
  // Set source data = 0x00, 0x01, ... ,0xFF
  b_arySramSrcData[i] = i & 0xFF;
  // Set destination data = 0x00
  b_arySramDstData[i] = 0x00;
}
```

After DMA is enabled, DMA starts transmitting from source SRAM address to destination SRAM address. The DMA TC flag and interrupt will be set when all data transfers are completed.

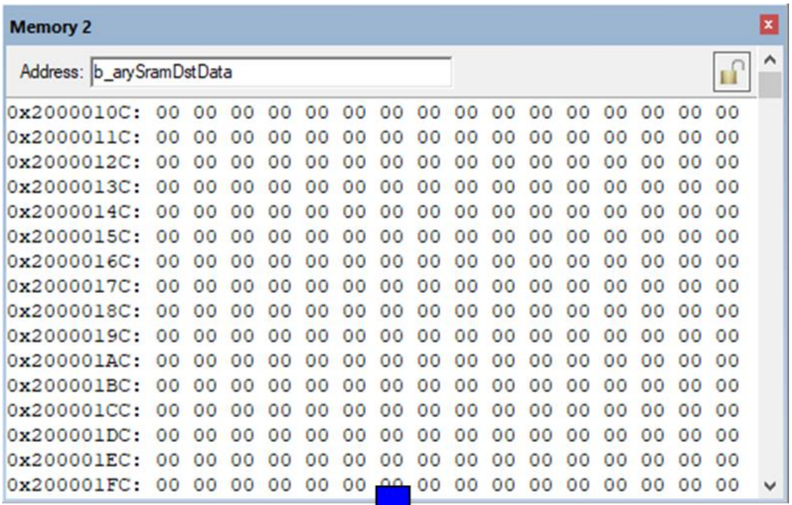
```
/* Example 2: DMA SRAM to SRAM Start*/

// Start DMA transfer
DMASramtoSramStart();

// Wait for DMA TC interrupt flag
while(stDMA0IntFlag[eDMA_CH0].Flag.bits.TC != 1);

// Check TX data = RX data
for(i = 0; i<DMA_TEST_SIZE ;i++)
{
  if(b_arySramSrcData[i] != b_arySramDstData[i])
    while(1); //Fail
}

/* Example 2: DMA SRAM to SRAM End*/
```

5 RESULT

The results can be observed through the Memory window.

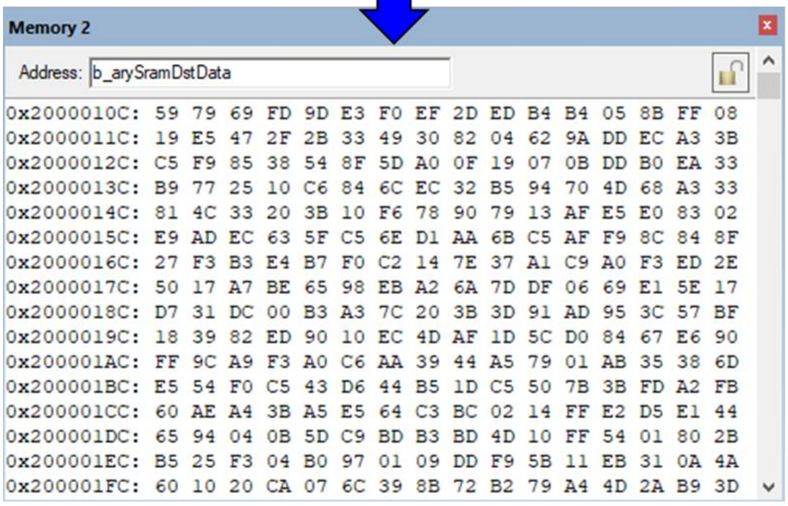
5.1 DMA FLASH to SRAM transfer

The data in the destination SRAM will be written as the data in source FLASH.



Memory 2
Address: b_arySramDstData

```
0x2000010C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000011C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000012C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000013C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000014C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000015C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000016C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000017C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000018C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x2000019C: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x200001AC: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x200001BC: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x200001CC: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x200001DC: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x200001EC: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x200001FC: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

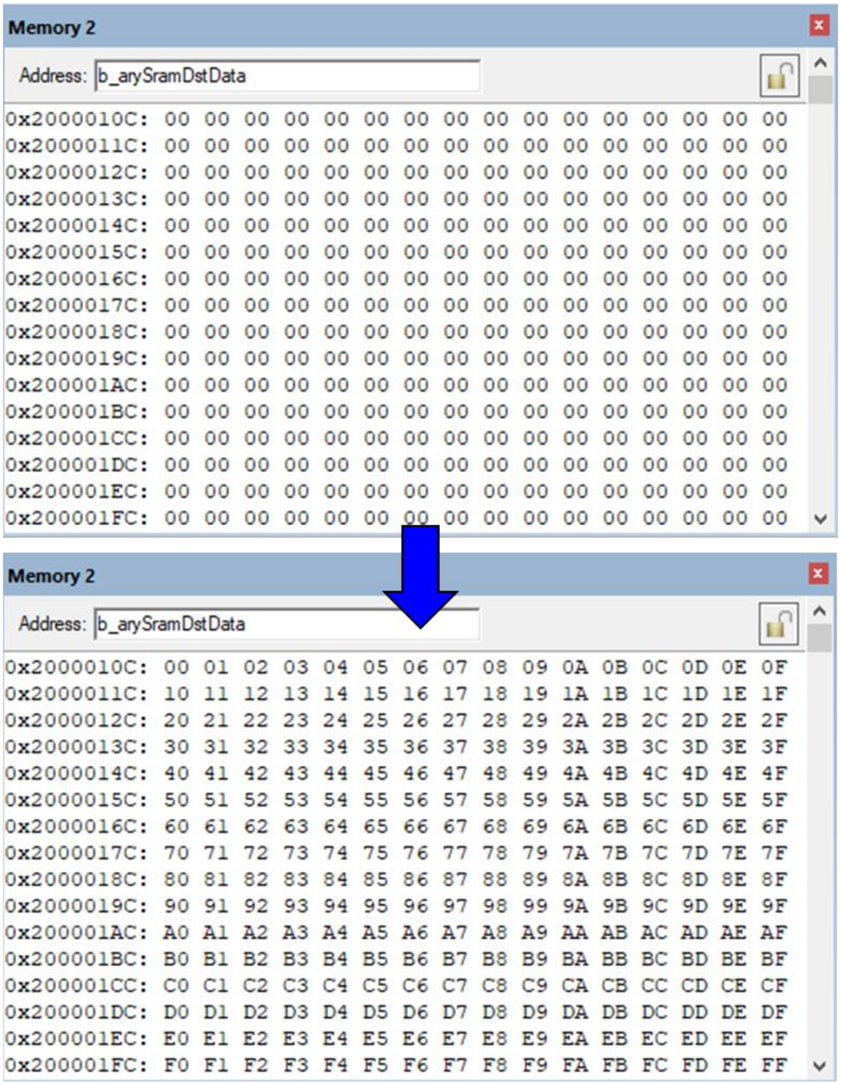


Memory 2
Address: b_arySramDstData

```
0x2000010C: 59 79 69 FD 9D E3 F0 EF 2D ED B4 B4 05 8B FF 08
0x2000011C: 19 E5 47 2F 2B 33 49 30 82 04 62 9A DD EC A3 3B
0x2000012C: C5 F9 85 38 54 8F 5D A0 0F 19 07 0B DD B0 EA 33
0x2000013C: B9 77 25 10 C6 84 6C EC 32 B5 94 70 4D 68 A3 33
0x2000014C: 81 4C 33 20 3B 10 F6 78 90 79 13 AF E5 E0 83 02
0x2000015C: E9 AD EC 63 5F C5 6E D1 AA 6B C5 AF F9 8C 84 8F
0x2000016C: 27 F3 B3 E4 B7 F0 C2 14 7E 37 A1 C9 A0 F3 ED 2E
0x2000017C: 50 17 A7 BE 65 98 EB A2 6A 7D DF 06 69 E1 5E 17
0x2000018C: D7 31 DC 00 B3 A3 7C 20 3B 3D 91 AD 95 3C 57 BF
0x2000019C: 18 39 82 ED 90 10 EC 4D AF 1D 5C D0 84 67 E6 90
0x200001AC: FF 9C A9 F3 A0 C6 AA 39 44 A5 79 01 AB 35 38 6D
0x200001BC: E5 54 F0 C5 43 D6 44 B5 1D C5 50 7B 3B FD A2 FB
0x200001CC: 60 AE A4 3B A5 E5 64 C3 BC 02 14 FF E2 D5 E1 44
0x200001DC: 65 94 04 0B 5D C9 BD B3 BD 4D 10 FF 54 01 80 2B
0x200001EC: B5 25 F3 04 B0 97 01 09 DD F9 5B 11 EB 31 0A 4A
0x200001FC: 60 10 20 CA 07 6C 39 8B 72 B2 79 A4 4D 2A B9 3D
```

5.2 DMA SRAM to SRAM transfer

The data in the destination SRAM will be written as the data in source SRAM.



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